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Classifying microfossil radiolarians on fractal pre-trained vision transformers

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While deep learning techniques, especially image classification using deep learning, continue to evolve, it has been noted that there is a large time gap in applying these techniques in geological studies. Recently, a new architecture called the vision transformer (ViT), which is an alternative to convolutional neural networks (CNN), has attracted considerable attention. In addition, it has been proposed that the pre-training of classification models using mathematically generated images instead of real images, called formula-driven supervised learning (FDSL), achieves a comparative or even higher performance in visual understanding. In this study, we applied these new techniques to the classification of microfossils (radiolarians). Compared with a previous CNN model, the ViT-based model achieved 6–8% higher average precision. On average, the precision of the FDSL pre-trained models was slightly higher than that of the models pre-trained on real images. Therefore, we propose that these techniques may be suitable for image classification in geological tasks.

Keywords Machine learning, Deep learning, Classification microfossils, Vision transformer, Formula-driven supervised learning

Microfossils found in oceanic and terrestrial sediments have long been studied as key indicators for estimating the depositional age and environment^{1–3}. They continue to play a crucial role in various fields, including resource exploration, understanding Earth's systems in response to past warming events, and uncovering the evolution of life. However, challenges such as the labour-intensive nature of observation and the difficulties in passing down technical expertise have persisted.

Knowledge in computer vision, especially deep learning, is accumulating exponentially. This has led to significant advances in various fields, including Earth Science^{4–6}. Machine-learning techniques have been applied to various palaeontological observations, such as microfossils^{1,3,7–13}. However, it has been pointed out that there has been a ten-year delay in the implementation of machine learning models in this field¹⁴, which may impede progress.

Despite the development of deep learning-based systems in microfossil classification^{7,8,12}, collecting images of rare classes (or species) remains difficult. Thus, a deep-learning architecture that can achieve good performance with a relatively small dataset is keenly required in this field. While most microfossil classification tasks adopt convolutional neural networks (CNN)^{7–9,11,13,15,16}, a new architecture called vision transformer (ViT) has been proposed¹⁷, and transformer-based models have achieved state-of-the-art performance in general image classification tasks¹⁸.

Pre-training is essential for achieving high classification accuracy with a relatively small number (< 10,000) of images¹⁹. Pre-training is often performed using datasets of diverse real images collected on the Internet, such as ImageNet²⁰ and COCO²¹. However, since these datasets include images of artificial substances such as vehicles and houses, it is unclear if they are suitable for classifying natural samples such as microfossils. Instead of using real images, Kataoka et al.²² proposed using mathematically generated images for pre-training, named “formula-driven supervised learning (FDSL)”. Fractal-like geometry is observed in many natural objects^{23,24}, which may reflect evolutionary adaptation²⁵. Therefore, it is expected that a higher classification accuracy can be obtained using a model pre-trained on mathematically generated images (Fig. 1).

In this study, we evaluated the applicability of these recent developments, ViT and FDSL, to microfossil studies by training image classification models on a dataset “SO32 (Supplementary material 2)”. SO32 dataset

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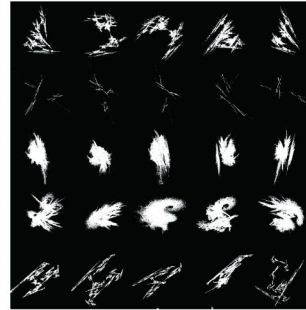
Pre-training

Real Images ImageNet (ImageNet-21k)

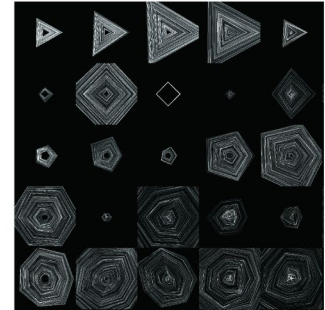


Formula-driven Images

Extended Fractal (ExFractal-21k)



Radial Contour (RCDB-21k)



Fine-tuning

SO32

actual microfossil images

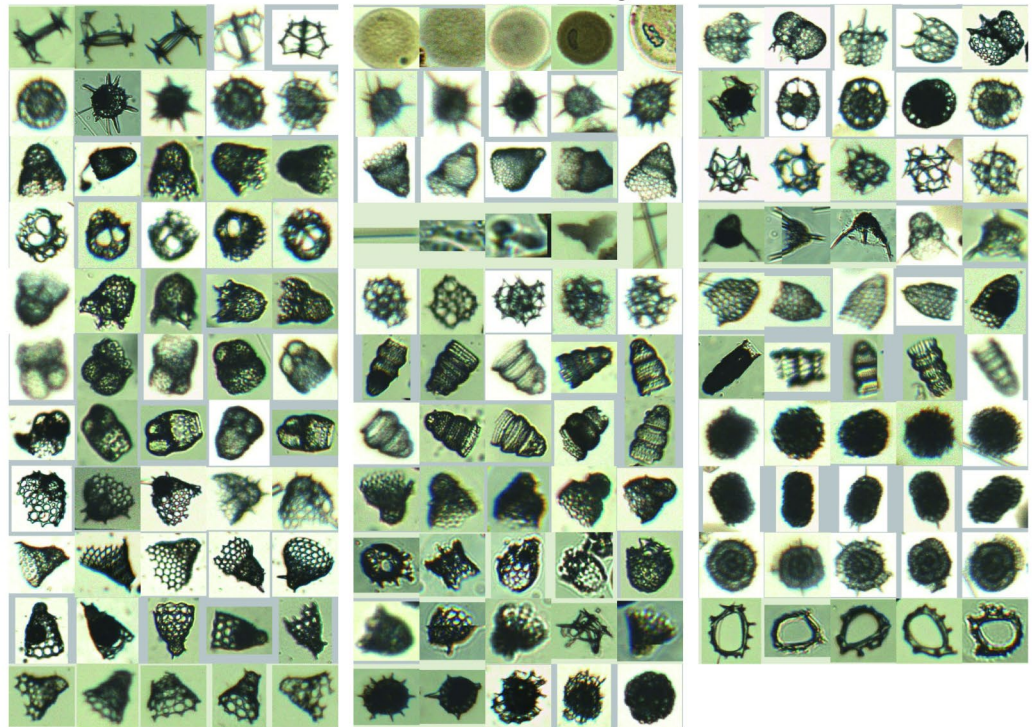


Fig. 1. Sample images of all the datasets considered in this study. Image classification models pre-trained on one real image dataset “ImageNet-21k”²⁰ and two formula-driven image datasets “ExFractal-21k” and “RCDB-21k”²² were fine-tuned using a radiolarian image dataset “SO32”.

consists of ~ 53,000 images of segmented microscopic images, which were categorized into 32 classes by a human annotator (Table 1). ViT models pre-trained on the real image dataset “ImageNet”²⁰ and two formula-driven datasets, extended fractal database (“ExFractal”) and radial contour database (“RCDB”) ²², were fine-tuned on the same dataset SO32. After training, the class-averaged accuracies were compared each other, in addition to that of CNN model pre-trained on a real image dataset by a previous study¹⁹. The detailed description of datasets and analytical procedure is in the methods section.

Class	Code	Number of images		
		All	Train	Test
<i>Antarctissa denticulata-cyrrindrica</i>	And	253	211	42
<i>Antarctissa longa-strelkovi</i>	Ast	290	242	48
<i>Antarctissa juvenile</i>	Ant	689	575	114
<i>Ceratocyrtis historicus</i>	Chs	62	52	10
<i>Cycladophora bicornis</i>	Cbc	307	256	51
<i>Cycladophora cornutoides</i>	Cct	219	183	36
<i>Cycladophora davisiana</i>	Cdv	9881	1000	1646
<i>Sethoconus tablatus</i>	Stb	32	27	5
<i>Eucyrtidium</i> spp.	Euc	40	34	6
<i>Lithocampe furcaspiculata</i>	Lfu	156	130	26
<i>Lithocampe platycephala</i>	Lpl	90	75	15
<i>Siphocampe arachnea</i> group	Sar	36	30	6
<i>Lithomelissa setosa-borealis</i>	Lst	346	289	57
<i>Lophophana</i> spp.	Lop	248	207	41
<i>Botryocampe antarctica</i>	Ban	244	204	40
<i>Botryocampe inflatum-conithorax</i>	Bin	51	43	8
<i>Phormospyris stabilis antarctica</i>	Psa	161	135	26
<i>Plectacantha oikiskos</i>	Poi	158	132	26
<i>Pseudodictyophimus gracilipes</i>	Pgr	170	142	28
<i>Acanthodesmia micropora</i>	Amp	74	62	12
<i>Zygocircus</i> spp.	Zyg	201	168	33
Other Nassellaria	OthN	194	162	32
<i>Actinomma leptoderma-boreale</i>	Alb	84	70	14
<i>Druppattractus irregularis-bensoni</i>	Dir	316	264	52
<i>Phortycium clevei-pylonium</i>	Ppy	237	198	39
<i>Larcoids</i> inner	Lar	205	171	34
<i>Spongurus pylomaticus</i>	Spy	107	90	17
<i>Spongodiscus</i> spp.	Spo	1132	944	188
<i>Sylodictya</i> spp.	Sty	416	347	69
Other Spumellaria	OthS	90	75	15
Diatoms	Dtm	11,115	1000	1852
Fragments	Frg	25,719	1000	4286

Table 1. Class names, codes, and numbers of images in the SO32 dataset.

Iteration	ViT ExFractal	ViT RCDB	ViT ImageNet	CNN ¹⁹
0	0.8661	0.8722	0.8605	0.782
1	0.8638	0.8606	0.8493	–
2	0.8419	0.8611	0.8293	–
3	0.8592	0.8713	0.8698	–
4	0.8338	0.8500	0.8714	–
Average	0.8529	0.8630	0.8561	0.782
Std	0.0128	0.0081	0.0155	–

Table 2. Averaged accuracies on five iterations. Mean and standard deviation of iterations are also shown.

Results and discussion

Impact of ViT and FDSL in classifying radiolarians

Table 2 shows averaged accuracies of five iterations. Figure 2 illustrates the average and one standard deviation, compared with conventional CNN-based training¹⁹. Compared with the conventional method, three ViT models trained in this study achieved 7.1–8.1% higher accuracies in 5-iteration average. We have also tested the performances of various CNN models but averaged accuracies were at the same level or lower than those of the three ViT-based models (Fig. S2). Thus, we suggest using ViT in microfossil classification tasks. It has been pointed out that ViT models, compared to CNN models, tend to focus more on the shape of objects rather than

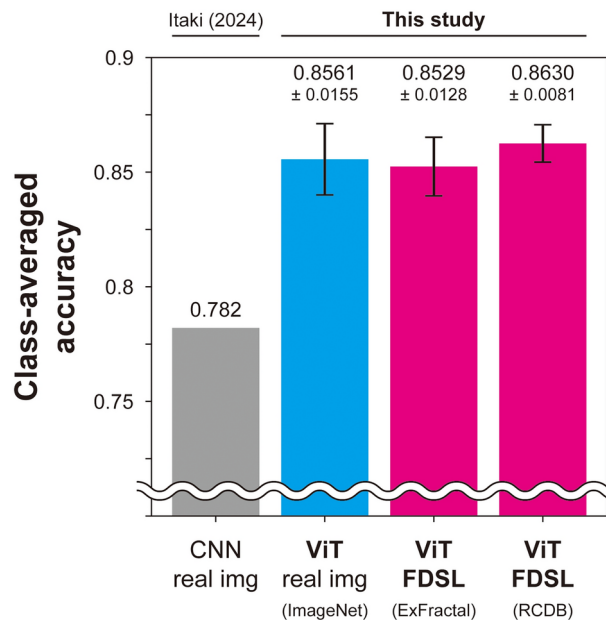


Fig. 2. Averaged accuracy of five iterations conducted in this study, compared to the results of CNN-based training¹⁹. The CNN and ViT models pre-trained on ImageNet have been treated as de-facto-standard models in computer vision field.

their texture²⁶, although CNN models also have the capability to learn shapes²⁷. The difference between ViT and CNN models in this study might be attributed to the history of human microfossil classification, which has relied largely on their shape rather than texture.

Averaged accuracies of the models pre-trained on mathematically generated (ExFractal, RCDB) and real images (ImageNet) were consistent within one standard deviation. However, RCDB pre-trained model achieved slightly higher accuracies on average than the ImageNet pre-trained model. This might be attributed to the relative similarity of RCDB to natural microfossils compared to ImageNet (Fig. 1). Although further study is needed, it is possible that the application of FDSL would yield higher performance in classification of natural objects such as microfossils.

Detailed view on the best model

The confusion matrix for the best model (RCDB pre-trained ViT, iteration 0) is shown in Fig. 3. We found that the model tended to show a higher classification accuracy as the number of images available for training increased. In particular, the accuracy for each class can be predicted by the equation $y \geq 1 - \frac{30}{x}$, where x and y represent number of training images and accuracy, respectively (Fig. 4). Using this approximation, it can be estimated that 100, 150, and 300 images for each class were desirable to achieve 70, 80, and 90% accuracy, respectively.

Two classes, *Lophophana* spp. (Lop) and other Nassellaria (OthN), showed accuracies that were significantly below the approximation curve. For class Lop, the model tended to make errors in predicting Lop as Ast (*Antarctissa longa-strelkovi*). Distinguishing between classes Lop and Ast is difficult, even for human observers (Fig. S3). The class OthN is a collection of shapes with fewer unique characteristics. More accurate classification of these difficult classes requires a larger amount of training data. However, it is still worth noting that a precise classification model can be trained from a relatively small number (several hundred) of images by applying the latest techniques.

Future implications

We used a vision transformer (ViT) and formula-driven supervised learning (FDSL) to classify radiolarians. Compared with the previously proposed CNN model, the ViT-based model achieved a higher average precision. The FDSL model achieved a slightly higher average precision than a model pre-trained on real images, although their differences were within the range of one standard deviation. Thus, we propose that these new techniques are worth applying to the classification of various microfossils and could provide new insights for developing related fields such as micropalaeontology, sedimentology, and oceanography.

Methods

Dataset used in this study

In this study, a dataset “SO32” with ~53,000 images (Supplementary material 2) was used. Glass slides were prepared from deep-sea sediment samples obtained from Integrated Ocean Discovery Program (IODP) Site U1422 (43° 5′ N, 138° 5′ E, 3429 m of water depth) in the Japan sea and DCR-1PC (45.7° S, 44.4° E, 2445 m of water depth) in the Southern Ocean. Particles contained in the slides were automatically segmented from microscopic images by the “miCRAD” system¹⁹, which is composed of a microscope with motorized XY stage,

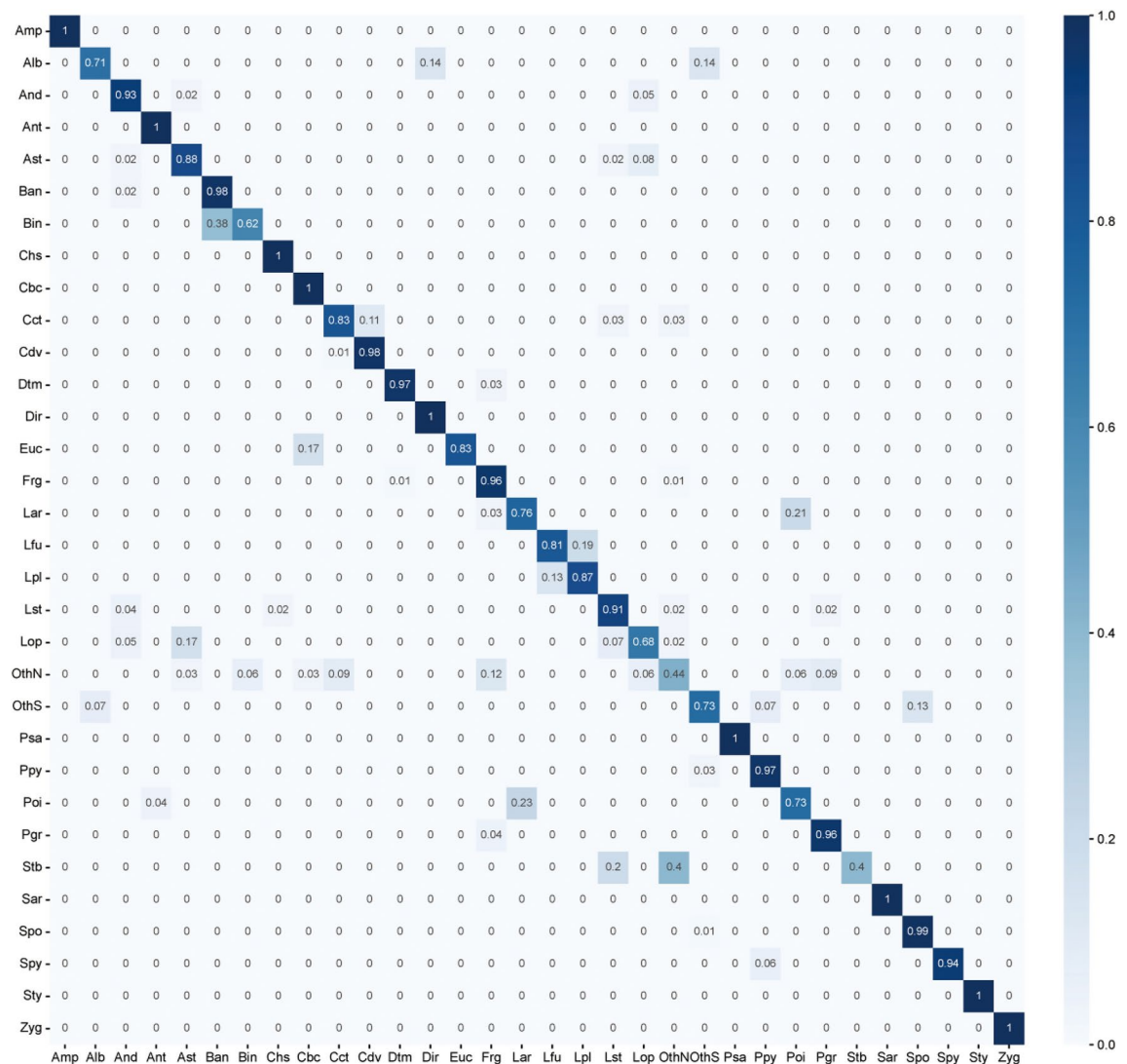


Fig. 3. Confusion matrix of radiolarian classification by RCDB pre-trained ViT model (iteration=0).

micromanipulator, a set of software that can concatenate tiled images, segment particles by thresholding and classify segmented particles using a deep learning model. Particles segmented by the system were re-classified into 32 classes (30 classes of radiolarians, one class for diatoms, and one class for fragments) by one of the authors (TI), who has studied radiolarians for more than 30 years.

Each segmented image was converted to 224×224 pixels without changing the aspect ratio. The images were then split into training (~85%) and testing (~15%) datasets. To avoid extreme class imbalance, the number of training images for classes with more than 1000 images was randomly reduced to 1000. The number of images and images for the training and test datasets are listed in Table 1.

Vision transformer

Vision Transformer (ViT) is a novel architecture for image recognition in the field of computer vision¹⁷ based on a transformer model that has achieved significant success in natural language processing²⁸. While traditional computer vision methods rely predominantly on CNN^{29,30}, ViT has opened new possibilities for image processing by applying the transformer approach.

The transformer is based on a self-attention mechanism, which can simultaneously consider the relationships between all the positions in the input data. This capability is widely used in natural language processing tasks to efficiently capture the relationships between words in sentences. In ViT, an image is first divided into small patches (e.g. 16×16). These patches were flattened and converted into feature vectors via linear embedding. Positional encoding, which carries the spatial information of the patches, is added to these embeddings to retain their spatial relationships before feeding them into the transformer. The transformer encoder receives the patch embeddings and learns their relationships using a self-attention mechanism. The final output is a global feature representation that integrates information from all of the patches, which is then input into a classification layer to predict the class of the entire image.

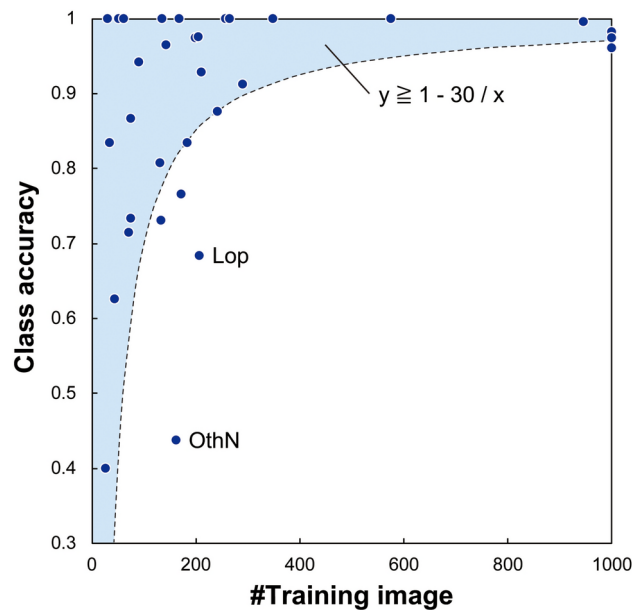


Fig. 4. Relationship between number of training images and accuracy for each class. Lop and OthN represent *Lophophana* spp. and other Nassellaria, respectively.

As of 2024, ViT-based architectures have achieved state-of-the-art accuracy in various tasks such as image classification¹⁸, object detection³¹, and instance segmentation³².

Formula-driven supervised learning

Formula-driven supervised learning (FDSL) is a new approach in image recognition related to pre-training²². Conventionally, pre-training of neural networks have been conducted on a vast amount of real image data, along with corresponding labels annotated by humans. However, collecting large datasets is costly and may cause ethical issues due to improper labelling.

FDSL uses image data automatically generated from mathematical formulas instead of real images (Fig. 1). Specifically, it employs fractal geometry, a mathematical technique for creating images, and simultaneously generates the corresponding labels. Fractal geometry is suitable for this purpose because it can produce highly complex patterns from simple equations, enabling the rapid and cost-effective creation of the large-scale image datasets necessary for pre-training. Because no real image is used, the new method also clears ethical issues. Compared to models pre-trained on real images, the FDSL model achieved a slightly higher performance in general image classification tasks²².

The previous study²² provided models pre-trained on two formula-driven datasets, an extended fractal database (ExFractal) and a radial contour database (RCDB). A dataset “ExFractal” is created by generating 3D self-similar, scale-invariant patterns using an iterated function system²³, and projecting them to 2D by randomly determined viewpoints. In contrast, RCDB is created by superimposing polygons radiating from a central point, with transformations such as resizing, and line width adjustments. These datasets provide complementary approaches for developing pattern recognition systems, with ExFractal emphasizing spatial complexity and RCDB targeting contour irregularity and diversity²².

Training and test protocol

A middle size ViT model, namely ViT-base which has 86.6 M parameters was used in this study. Three different pre-trained models with real images (“ImageNet-21 k”) and mathematically generated images (“ExFractal-21 k” and “RCDB-21 k”) were fine-tuned using a training subset of the SO32 dataset. When the images were loaded during training, they were randomly flipped vertically and horizontally for online augmentation. The cross-entropy weighted by the inverse of the number of samples per class was used as the loss function. The initial learning rate was set at 0.001, and stochastic gradient descent with a momentum of 0.95 was used as an optimiser. Forty training epochs were conducted. After training each epoch, the performance of the model was evaluated using the test dataset. The test loss remained almost constant after 20 epochs (Fig. S1). The best epoch was determined using the minimum loss in the test subset. Using the best epoch model, the classes of the test images were predicted to generate a confusion matrix.

To measure whether features were learned for all classes from the imbalanced dataset, class-averaged accuracy, which is calculated by the following steps, was used as a metric of the model performance. First, accuracy for class i is calculated as follows:

$$Accuracy_i = \frac{(\text{number of test images for class } i \text{ which is correctly predicted by a model})}{(\text{number of test images for class } i)} \quad (1)$$

Then, class-averaged accuracy is an average of accuracies for all the classes.

$$\text{Class_averaged_accuracy} = \frac{\sum_i \text{Accuracy}_i}{(\text{number of classes})} \quad (2)$$

To evaluate the uncertainties due to the initial conditions, the models were trained five times with different random seeds for each condition. The program codes and pre-trained weights are available on GitHub (<https://github.com/masora1030/CVPR2022-Pretrained-ViT-PyTorch>).

Data availability

The radiolarian dataset “SO32” used in this study is available in supplementary material 2. The program codes are available on GitHub (<https://github.com/masora1030/CVPR2022-Pretrained-ViT-PyTorch>).

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Author contributions

K.M.: conceptualization, investigation, writing—original draft, funding acquisition. T.I.: conceptualization, resources, writing—review and editing, funding acquisition. H.K.: methodology, software, writing—review and editing. A.M.: investigation, writing—review and editing.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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